

Elements and atoms

EVERYTHING IS BUILT FROM ELEMENTS AND ATOMS.

In ancient times, people believed that our world was made from just a few elemental substances: earth, air, fire, and water. Chemists now know that it is made from 90 naturally occurring elements.

What is an element?

An element is a substance that cannot be broken down into simpler constituents. Therefore a pure sample of an element is made entirely from one type of atom. The structure of that atom defines the element's physical and chemical properties.

Atomic number
Every element has a unique atomic number, which is the total number of protons in the atom.

1 1.0079
H
HYDROGEN

Atomic mass
Atomic mass is the mass of protons, neutrons, and electrons in the atom. The number is shown as an average of all isotopes (see pages 111 and 169).

26 55.845
Fe
IRON

3 6.941
Li
LITHIUM

17 35.453
Cl
CHLORINE

Chemical symbol
Every element is abbreviated using a unique symbol of one or two letters. Mostly these relate to the English names, so H is for hydrogen and Cl is for chlorine. A few are based on other languages, for example iron is Fe, from the Latin word "ferrum."

Name
Many element names are very old. Newer ones are agreed by an international committee. Chlorine is named after khloros, the Greek word for "greenish yellow" (the gas's color).

SEE ALSO

◀ 78–79 Cycles in nature

Octet rule 112 ▶

Periodic table 116–117 ▶

Size of atoms 118 ▶

Inner and outer electrons 124 ▶

Inside atoms 168–169 ▶

REAL WORLD

Hennig Brand

The German Hennig Brand is the first historical figure known to have discovered a new element. In 1669, he found phosphorus after investigating the substances in his urine. The phosphorus glowed in the dark, making Brand think he had found a magic material.



Atomic structure

An atom is made up of positively charged protons and negatively charged electrons. The atoms of every element have a specific number of protons. Protons have a positive charge, but atoms are always neutral because the protons are balanced by an equal number of negatively charged electrons.

Proton
A proton is a positively charged particle that sits in the nucleus.

Electron shell
The electrons are arranged in shells, or energy levels, around the nucleus.

Nucleus
All atoms, except hydrogen atoms, have neutrons as well as protons in their nucleus.

Neutrons
Neutrons are neutral particles that have no charge.

Electrons
An electron is a negatively charged particle that sits in the electron shell.

1 H △ **Hydrogen**
With one proton and one electron, hydrogen atoms are the smallest, lightest, and simplest of any element.

7 N △ **Nitrogen**
Nitrogen atoms have seven protons and seven electrons. Most nitrogen atoms also have seven neutrons.